

# MATERIAL AND ENERGY BALANCES

CHE 210, CREDIT HOURS: 4  
FALL 2025

## I. INSTRUCTOR & COURSE DETAILS

INSTRUCTOR: LEWIS E. WEDGEWOOD

Email address: [wedge@uic.edu](mailto:wedge@uic.edu)

Drop-In Office Hours (in-person and virtual): Mondays & Wednesdays 2:00 PM-3:00 PM

Drop-In Hours location: EIB 250

Join URL: <https://uic.zoom.us/j/8484146077>

## TEACHING ASSISTANTS

BRYCE ASKEW

Drop-In Office Hours: TBD

Drop-In Hours location: EIB TBA

E-mail: [baske@uic.edu](mailto:baske@uic.edu)

## COURSE BLACKBOARD SITE

<https://uic.blackboard.com>

Students are expected to log into the course site regularly to learn about any developments related to the course, upload assignments, and communicate with classmates. For all technical questions about Blackboard, email the Learning Technology Solutions team at [LTS@uic.edu](mailto:LTS@uic.edu).

## COURSE MODALITY AND SCHEDULE

This course is taught ON CAMPUS and design for in-person participation.

Lectures: Mondays and Wednesdays 12:30 AM - 1:45 PM

Discussion: Fridays 12:30 AM - 1:45 PM

EIB Room 124

## CALENDAR

|             |                                   |
|-------------|-----------------------------------|
| -- 09/01    | Labor Day (No Classes)            |
| -- 10/01    | First Midterm Exam                |
| -- 11/05    | Second Midterm Exam               |
| -- 11/26    | Wellness Day                      |
| -- 11/27-28 | Thanksgiving Holiday (No Classes) |
| -- 12/09    | FINAL EXAM (8 AM – 10 AM)         |

## II. COURSE INFORMATION

### CATALOG COURSE DESCRIPTION AND PREREQUISITE/COREQUISITE STATEMENT

Material and energy balances applied to chemical systems. Introduction to chemical and physical properties. Introduction to the use of computers for chemical process calculations. Course Information: Prerequisite(s): Credit or concurrent registration in [CHE 201](#); and credit or concurrent registration in [CHE 205](#).

This course is a continuation of the material covered in ChE 201 Thermodynamics. In that previous course, you became familiar with the concepts of energy and entropy balances for pure components systems. Also, we will make use of the material covered in ChE 205 for solving problems. In ChE 210, we will study more complex systems including mixtures and reacting systems. We will also include more complex processes with several feed and effluent streams. The chemical processes will also contain unit operations such as separators and reactor. This increased level of complexity will require more sophisticated methods of analysis. We will also begin to rely more on numerical methods to help us solve these more challenging problems. You will become familiar with some basic numerical methods and applications.

This course lays out the general approach of chemical engineering. Later courses will deal in more detail with many of the particulars discussed in ChE 210. For instance, Mass Transfer Processes ChE 313 treats separation processes in great detail, Reactor Design ChE 321 studies the engineering concepts in industrial-scale reactors, and Process Design ChE 396/7 investigates the creation of new and better processes. If a student can master the fundamentals in ChE 210, this knowledge will be of benefit throughout the rest of the curriculum as well as in a professional engineering career.

It is assumed that the students in the course have a thorough grounding in elementary mathematics, and this mathematical background will be used; some mathematics review may be necessary at various points in the course. By the end of the term the students should understand material and energy concepts; they should be able to apply control volume analysis to a variety of situations in order to solve elementary chemical engineering process problems.

The analysis of chemical processes is conceptually difficult and requires time and effort on the student's part before it can be well understood. Students should read the appropriate sections of the text before lectures, homework assignments should be started as soon as possible, and thoughtful questions should be brought up during lecture and discussion. Good luck!

### COURSES LEARNING OUTCOMES

At the completion of this course, students should be able to:

- Calculate a simple economic analysis of a chemical process. Evaluate the atomic and process efficiency of a chemical process.
- Formulate material and energy balances on chemical processes represented in block flow diagrams.
- Understand the essential physical principles of material and energy balances as applied to chemical processes.

- Set up differential material and energy balances and analyze the degrees of freedom. Students should be able to solve problems when they are well posed and appropriate analyze problems that are over or under specified.
- Solve sets of algebraic equations, linear or nonlinear. Also, solve first-order separable differential equations for transient balance equations.
- Use thermodynamic data to solve energy balances on multicomponent, reacting multi- unit operations processes.

#### ABET STUDENT LEARNING OUTCOMES CENTRAL TO THE COURSE

- ABET Student Outcome 1: an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- ABET Student Outcome 2: an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- ABET Student Outcome 7: an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

#### BRIEF LIST OF TOPICS TO BE COVERED

##### I. Overview of the Chemical Industry

- A. Applications of Mass and Energy Balance in Chemical Processes
- B. The Solvay Process
- C. The Haber Process

##### II. Applications of Mathematics

- A. Solving Systems of Algebraic Equations
- B. Fitting Experimental Data
- C. Find Roots to Nonlinear Equations

##### III. Mass Balances

- A. Process Flow Diagrams
- B. Multicomponent Systems with a Single Unit Operation
- C. Systems with Chemical Reaction
- D. Systems with Multiple Unit Operations
- E. Systems with Recycle and Bypass Streams
- F. Unsteady Processes
- G. Extent of Reaction
- H. Equilibrium Reactions
- I. Conversion, Yield, and Selectivity

##### IV. Energy Balances

- A. Enthalpies of Reaction
- B. Hess's Law – Hypothetical Thermodynamic Pathways
- C. Reactor Heat Duty
- D. Thermodynamic Properties – The NIST Chemistry Webbook

- E. Large Processes with Chemical Reaction and Multiple Unit Operations
- F. Combined Mass and Energy Balances

## BRIEF LIST OF TEXTBOOK READING ASSIGNMENTS

### Appendix A – All Sections

- (i) Linear algebra
- (ii) Roots of nonlinear algebraic equations
- (iii) Fitting data

### Chapter 1 – All Sections

### Chapter 2 – All Sections

### Chapter 3 – All Sections

We will not be using the integral form of the material balance. These equations have an assumption that restricts their application. Instead, we will always use the differential material balance equations which can be applied generally.

### Chapter 4 – All Sections

In Section 4 of Chapter 4 students will not be required to calculate equilibrium constants  $K_a$  from thermodynamic data. This material is covered in ChE 301. Instead, in ChE 210  $K_a$  will be supplied. It may depend on temperature and pressure.

### Chapter 5 – Only Sections 1 - 3

### Chapter 6 – All Sections

We will not cover the Engineering Bernoulli Equations which is covered in ChE 311.

## COURSE MATERIALS

### REQUIRED TEXTBOOK

***Introduction to Chemical Processes: Principles, Analysis, Synthesis, Murphy, 2007 1<sup>st</sup> Edition***

by Regina Murphy

Note: Do not buy the 2<sup>nd</sup> edition!

## III. COURSE POLICIES & CLASSROOM EXPECTATIONS

### GRADING POLICY AND POINT BREAKDOWN

- Discussion: 5%
  - Homework: 10% Not scoring at least half of the total homework grade is grounds for failing the course.
  - Quizzes: 15%
  - Midterm Exam I: 20%
  - Midterm Exam II: 20%
  - Final Exam 30%
- There is no "extra" credit.

## ASSIGNMENTS

- Reading assignments: Assigned from the textbook. No submission is required.
- Problem sets (Homework): Please submit an electronic copy of your homework by the specified due date before the start of the class.

## POLICY FOR MISSED OR LATE WORK

- Homework and Term Project: Late submissions will not be considered for partial credit. Unreadable and unorganized homework will not be graded.
- Quizzes: No makeup will be administered for the missed quizzes, however, one quiz with lowest grade will not be counted towards your final grade.
- Exams: Students who must miss an exam for any reason are expected to contact the instructor before the date of the exam.

## ATTENDANCE AND PARTICIPATION POLICY

Please email me if you face an unexpected situation that may impede your attendance, participation in required class and exam sessions, or timely completion of assignments.

## EMAIL EXPECTATIONS

Students are responsible for all information instructors send to your UIC email and Blackboard accounts. Faculty messages should be regularly monitored and read in a timely fashion.

## ACADEMIC INTEGRITY

As a student and member of the UIC community, you are expected to adhere to the [Community Standards of academic integrity](#), accountability, and respect. Please review the [UIC Student Disciplinary Policy](#) for additional information.

## OTHER COURSE POLICIES

Regrading of exams must be requested within 2 days after the grades are returned. For homework issues, please contact the TA.

## IV. ACCOMMODATIONS

### DISABILITY ACCOMMODATION PROCEDURES

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at [drc@uic.edu](mailto:drc@uic.edu), via email at [drc@uic.edu](mailto:drc@uic.edu), or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodation, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.

### RELIGIOUS ACCOMMODATIONS

Following [campus policy](#), if you wish to observe religious holidays, you must notify me by the tenth day of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify me at least five days before you will be absent. Please submit [this form](#) by email with the subject heading: "YOUR NAME: Requesting Religious Accommodation."

## V. CLASSROOM ENVIRONMENT

### INCLUSIVE COMMUNITY

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If aspects of this course result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify me as soon as possible.

### NAME AND PRONOUN USE

If your name does not match the name on my class roster, please let me know as soon as possible. I welcome your pronouns if you would like to share them with me. For more information about pronouns, see this page: <https://www.mypronouns.org/what-and-why>.

### COMMUNITY AGREEMENT/CLASSROOM CONDUCT POLICY

- Be present by turning off cell phones and removing yourself from other distractions.
- Be respectful of the learning space and community. For example, no side conversations or unnecessary disruptions.
- Use preferred names and gender pronouns.
- Assume goodwill in all interactions, even in disagreement.
- Facilitate dialogue and value the free and safe exchange of ideas.
- Try not to make assumptions, have an open mind, seek to understand, and not judge.
- Approach discussion, challenges, and different perspectives as an opportunity to "think out loud," learn something new, and understand the concepts or experiences that guide other people's thinking.
- Debate the concepts, not the person.
- Be gracious and open to change when your ideas, arguments, or positions do not work or are proven wrong.
- Be willing to work together and share helpful study strategies.
- Be mindful of one another's privacy, and do not invite outsiders into our classroom.

### CONTENT NOTICES AND TRIGGER WARNINGS

Our classroom provides an open space for a critical and civil exchange of ideas, inclusive of a variety of perspectives and positions. Some readings and other content may expose you to ideas, subjects, or views that may challenge you, cause you discomfort, or recall past negative experiences or traumas. I intend to discuss all subjects with dignity and humanity, as well as with rigor and respect for scholarly inquiry. If you would like me to be aware of a specific topic of concern, please email or visit my Student Drop-In Hours.

## VI. RESOURCES: ACADEMIC SUCCESS, WELLNESS, AND SAFETY

We all need the help and the support of our UIC community. Please visit my office hours for course consultation and other academic or research topics. For additional assistance, please contact your assigned college advisor and visit the support services available to all UIC students.

#### ACADEMIC SUCCESS

- UIC [Tutoring Resources](#)
- College of Engineering [tutoring program](#)
- [Equity and Inclusion in Engineering Program](#)
- [UIC Library](#) and [UIC Library Research Guides](#)
- [Offices](#) supporting the UIC Undergraduate Experience and Academic Programs.
- [Student Guide for Information Technology](#)

#### WELLNESS

- Counseling Services: You may seek free and confidential services from the Counseling Center at <https://counseling.uic.edu/>.
- Access [U&I Care Program](#) for assistance with personal hardships.
- Campus Advocacy Network: Under Title IX, you have the right to an education free from any form of gender-based violence or discrimination. To make a report, email [TitleIX@uic.edu](mailto:TitleIX@uic.edu). For more information or confidential victim services and advocacy, visit UIC's Campus Advocacy Network at <http://can.uic.edu/>.

#### SAFETY

- [UIC Safe App](#)—PLEASE DOWNLOAD FOR YOUR SAFETY!
- [UIC Safety Tips and Resources](#)
- [Night Ride](#)
- [Emergency Communications](#): By dialing 5-5555 from a campus phone, you can summon the Police or Fire for any on-campus emergency. You may also set up the complete number, (312) 355-5555, on speed dial on your cell phone.